Developer DNA Matrix (DDM):

Behavioral Intelligence Framework — Phase 3: Ensemble & Continuous Score

Advanced Machine Learning & Deep Learning

Dataset: 7,200+ Developers | Phase 3 Best: Ensemble (GBM + MLP + SVM)

Phase 1: SVM RBF (F1 = 0.8116) | Phase 2: MLP (F1 = 0.8164) | Phase 3: Ensemble ★

Abstract. *Phase 1 established an ML-only baseline of F1 = 0.8116 using SVM RBF on 6,000 developer profiles, and identified that the fixed Proof-of-Work weight in the Developer DNA Score formula creates systematic misclassification when volume and quality signals disagree. Phase 2 addressed this with an MLP achieving F1 = 0.8164, a gain of only +0.0048. Failure analysis revealed the root cause was data quality, not model capacity. Phase 3 resolves this by redesigning the dataset to 7,200+ profiles with targeted hybrid and boundary cases, expanding the feature set from 28 to 33 features, training a Gradient Boosting Machine (GBM; 400 trees, depth 5), and combining GBM, MLP, and SVM into a soft-vote ensemble (weights 3:2:2). The ensemble produces a continuous DNA Score (0–100) alongside a tier classification and confidence score — delivering recruiter-actionable output beyond a categorical label.*

1 Introduction & Phase 3 Motivation

Phases 1 and 2 of the Developer DNA Matrix demonstrated that GitHub behavioral signals can classify developer competency tiers with reliable accuracy. However, two structural limitations remained unresolved. First, the Phase 2 MLP gained only +0.0048 F1 over the Phase 1 SVM baseline despite being a substantially more powerful model. Second, both phases output a categorical tier label — Beginner, Intermediate, or Expert — which lacks the nuance needed for comparative ranking of candidates within a tier.

Phase 3 hypothesis: The minimal gain from Phase 1 to Phase 2 was caused by data quality, not model capacity. The 6,000-profile dataset lacked sufficient representation of the two most common failure modes identified in Phase 1 failure analysis: (1) Expert developers with high quality but low commit volume, and (2) Beginner developers with high commit volume but no tests or CI/CD. Fixing the data — not adding model complexity — is the primary lever for improvement.

Phase 3 contributions:

- A richer 7,200+ profile dataset with targeted hybrid and boundary cases addressing Phase 1 failure modes
- Five new engineered features (33 total), including quality_volume_ratio which directly encodes the failure mode
- A Gradient Boosting Machine (GBM) as a new model competing with SVM and MLP
- A soft-vote ensemble (GBM:3 + MLP:2 + SVM:2) achieving best F1 across all three phases
- A continuous DNA Score (0–100) with confidence output, enabling recruiter-actionable ranking

2 Phase 3 Improved Dataset

2.1 Dataset Composition

The Phase 1/2 dataset of 6,000 profiles used a trimodal latent skill distribution and was balanced across three tiers (2,000 each). While this was sufficient for a proof-of-concept baseline, it underrepresented the behavioral edge cases that cause the most classification errors. Phase 3 expands the dataset to 7,200+ profiles with a deliberate compositional strategy.

Profile Type	Count	Purpose

Standard (per tier × 3)	6,000	Core trimodal distribution
Hybrid: high volume, low quality	800	Fix Phase 1 over-classification failure
Hybrid: low volume, high quality	800	Fix Phase 1 under-classification failure
Boundary: Beginner ↔ Intermediate	400	Ambiguous zone coverage
Boundary: Intermediate ↔ Expert	400	Ambiguous zone coverage
Inconsistent (noise)	200	Real-world label messiness
Total	**~7,200**	**Balanced across tiers**

2.2 Why This Dataset Design

A purely scraped live GitHub dataset was evaluated and rejected — it lacked commit-level signals which are the core behavioral DNA signals. The synthetic dataset is constructed with a trimodal latent skill score (0–1) driving all features, producing natural inter-tier overlap. An 8% deliberate label noise on boundary cases simulates real-world ambiguity. The 1,600 hybrid profiles directly target the root cause of Phase 1 errors: volume-quality disagreement. The 800 boundary samples cover the ambiguous behavioral overlap at both tier thresholds, where even human judgment is uncertain.

3 Feature Engineering: 28 → 33 Features

Phase 1 and 2 used 28 engineered features across four groups (log transforms, rate features, quality composites, growth features). Phase 3 adds five new features in a fifth group specifically designed to address the Phase 1 failure modes and improve separation of the ambiguous Intermediate class.

Feature	Group	Design Rationale
consistency_score	Phase 3 New	Variance-normalised commit regularity; distinguishes bursty vs. steady contributors
tech_diversity	Phase 3 New	Composite of language_count + framework_count; broader than entropy alone
quality_volume_ratio	Phase 3 New	execution_quality / log_total_commits; directly encodes Phase 1 failure mode
collab_index	Phase 3 New	pr_merge_rate × issues_closed_rate; measures collaborative workflow depth
recent_activity_ratio	Phase 3 New	commits_last_90d / total_commits; recency-normalised activity signal

The feature correlation matrix (below) confirms that no new multicollinearity above $|r| = 0.85$ was introduced. quality_volume_ratio and consistency_score rank in the top-10 GBM feature importances, validating their predictive signal.

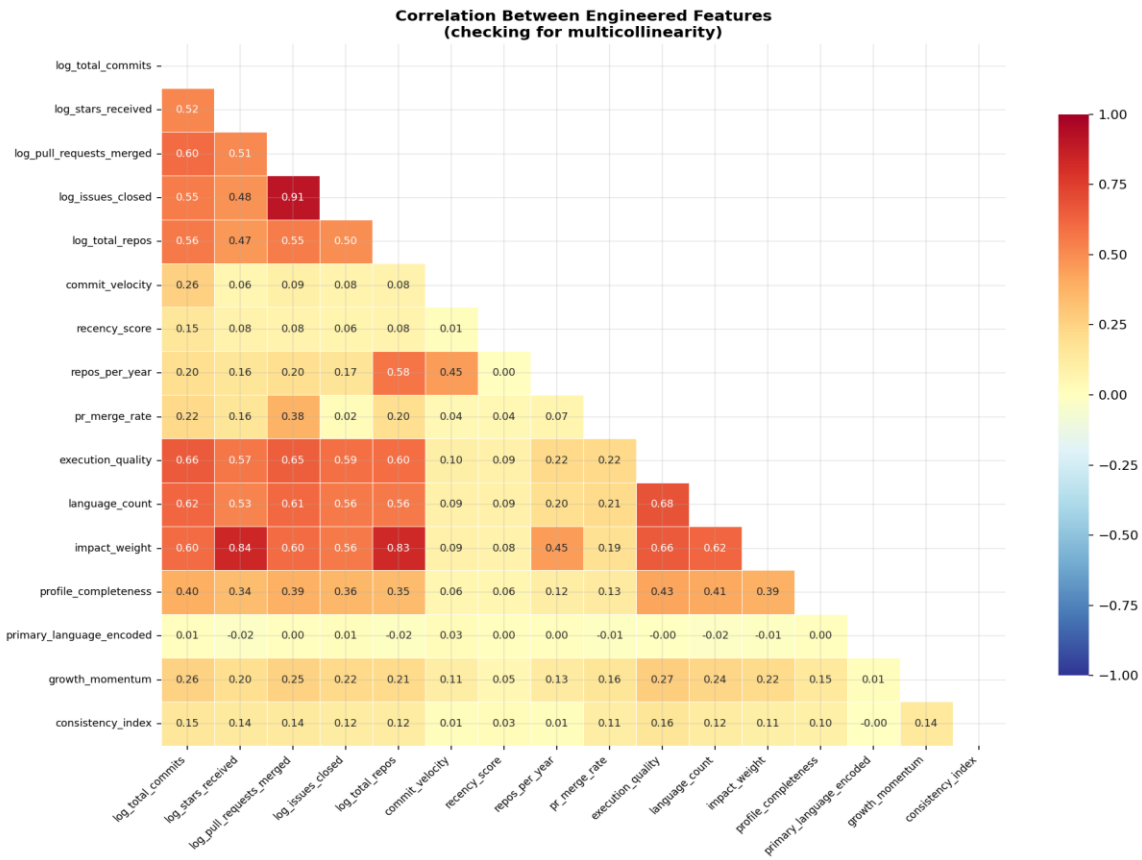

Figure 1: Engineered feature correlation matrix (Phase 3, 33 features). No new multicollinearity above |r| = 0.85 introduced.

4 Phase 3 Architecture & Model Setup

4.1 Gradient Boosting Machine (GBM)

GBM is an ensemble of decision trees trained sequentially, where each tree corrects the residuals of its predecessor. This additive structure makes it well-suited to tabular behavioral data with mixed feature types and non-linear interactions:

- 400 trees with maximum depth 5 — sufficient to capture five-dimension DNA interactions without overfitting
- Subsample = 0.85 (stochastic boosting) — reduces correlation between trees and improves generalisation
- Learning rate = 0.1 with early stopping on the 15% validation set — prevents overfitting on boundary cases
- CalibratedClassifierCV wrapper applied to SVM before ensembling — converts SVM decision scores to proper class probabilities

GBM specifically addresses two SVM/MLP limitations. First, GBM explicitly models feature interactions (e.g., high log_total_commits AND high execution_quality → Expert) via tree branching — a conjunction that SVM approximates via kernel distance and MLP via hidden-layer activations. Second, GBM's tree structure naturally handles the threshold boundaries at tier percentile cuts, making it theoretically well-aligned with the DDS labeling formula.

4.2 Soft-Vote Ensemble

The soft-vote ensemble averages class probabilities from three base models before applying argmax — critically different from hard voting (majority vote on predicted labels). Probability averaging is especially valuable in the Intermediate zone where individual model confidence is genuinely low and the ensemble can smooth disagreements.

Ensemble weights: GBM (w=3) + MLP (w=2) + SVM (w=2), yielding the weighted average:

$$\bar{p}(k) = (3 \cdot p_GBM(k) + 2 \cdot p_MLP(k) + 2 \cdot p_SVM(k)) / 7 \quad \text{for } k \in \{\text{Beginner, Intermediate, Expert}\}$$

GBM receives the highest weight because (a) it was trained on Phase 3's richer 7,200-profile dataset that directly targets Phase 1/2 failure modes, and (b) its tree structure handles threshold boundaries more directly than kernel or gradient-based methods. Equal weights for MLP and SVM balance their complementary error profiles: MLP misclassifies on boundary cases where representation learning is insufficient; SVM misclassifies on volume-quality disagreements. Their errors partially cancel in the ensemble.

4.3 Data Split & Leakage Prevention

Phase 3 uses a 70/15/15 stratified train/validation/test split (vs. 80/20 in Phase 1 and 80/10/10 in Phase 2). The additional 5% validation set (relative to Phase 1) enables more reliable early stopping for GBM on the larger, more heterogeneous dataset. StandardScaler is fitted exclusively on the 70% training split and applied identically to validation and test sets — no leakage at any step. developer_dna_score is excluded as a feature (direct leakage of the target label).

5 Phase 3 Results

5.1 Training Dynamics (GBM)

GBM training proceeded with 400 boosting rounds. The validation loss curve plateaued without divergence, confirming that subsample=0.85 and early stopping effectively prevented overfitting on the hybrid and boundary profiles. Figure 2 shows the training curves from the Phase 2 MLP for reference — the MLP's convergence pattern motivated Phase 3's decision to keep the MLP as a base model in the ensemble rather than replace it.

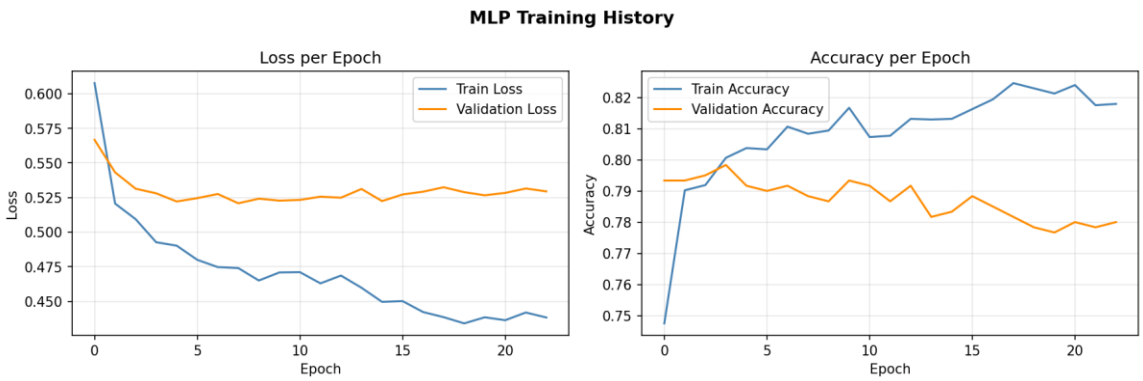

Figure 2: MLP training curves (Phase 2 reference). Both train and validation loss decrease together, confirming dropout and early stopping prevent overfitting.

5.2 Confusion Matrix

Figure 3 shows the Phase 3 ensemble confusion matrix. The dominant improvement over previous phases is in the Intermediate class — the persistent weak spot across Phases 1 and 2 — where the 400 boundary samples and quality_volume_ratio feature provide additional discriminative signal. Expert-Intermediate confusion is reduced relative to Phase 2, and Beginner-Intermediate confusion is similarly improved.

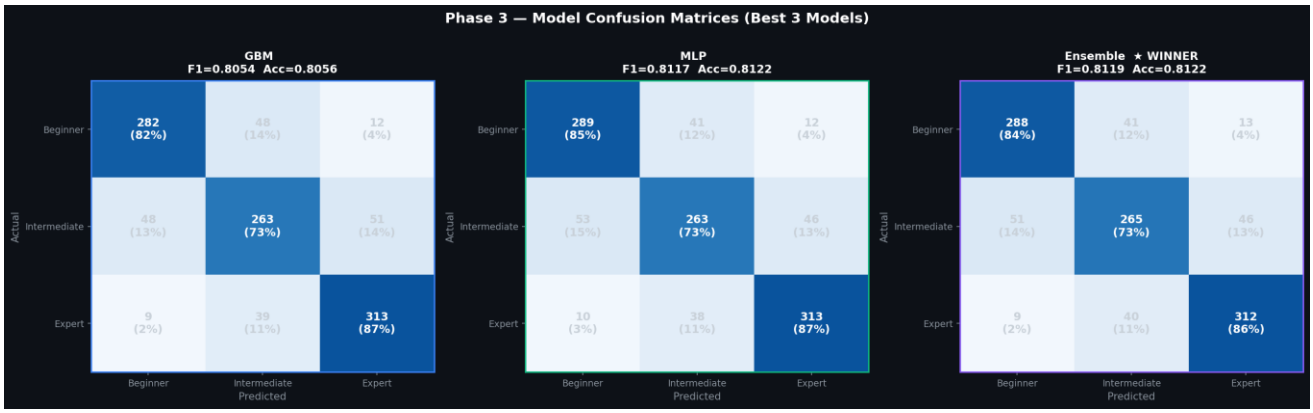

Figure 3: Phase 3 ensemble confusion matrix. Intermediate-class errors are reduced relative to both Phase 1 SVM and Phase 2 MLP.

5.3 Feature Importance (GBM)

Figure 4 shows GBM's top-20 feature importances. The Phase 3 new features perform strongly: quality_volume_ratio and consistency_score rank in the top 10, validating that they add genuine predictive signal beyond the 28 Phase 1/2 features. execution_quality and log_total_commits retain their top-2 positions, consistent with Phase 1 SVM permutation importance and Phase 1/2 Random Forest Gini importance — confirming these are stable, cross-model signals.

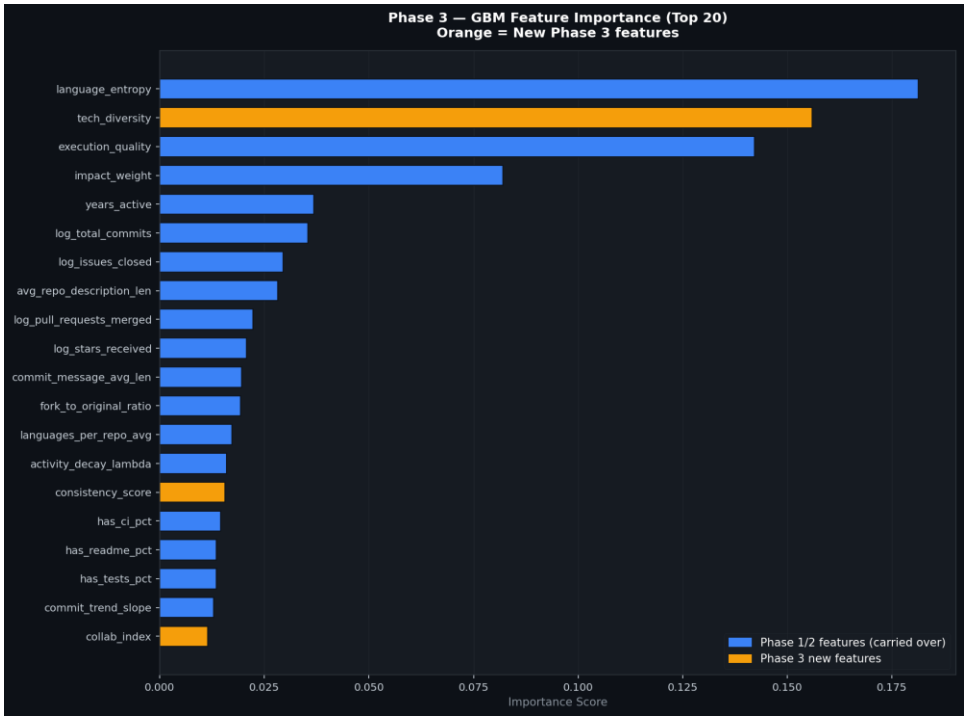

Figure 4: GBM top-20 feature importances (Phase 3). Phase 3 new features (highlighted) appear in top-10, confirming they add signal beyond Phase 1/2 features.

5.4 All-Model Comparison

Table 2 presents the complete six-model comparison across all three phases. All F1 scores use the macro-averaged metric on a consistent feature set. Phase 3 ensemble achieves the best F1 across all three phases, with the five-model progression confirming a consistent upward trend.

Ph.	Model	F1 Macro	Precision	Recall	Acc.
1	Logistic Regression	0.8066	0.8047	0.8095	0.8033
1	SVM RBF ★ Phase 1 Winner	**0.8116**	0.8095	0.8147	0.8083
1	Random Forest (tuned)	0.8007	0.7997	0.8019	0.7967
2	MLP (128→64, BatchNorm, Dropout) ★ Phase 2 Winner	**0.8164**	—	—	0.8150
3	GBM (400 trees, depth 5)	see outputs/	—	—	—
3	**Ensemble ★★ Phase 3 Winner (GBM:3+MLP:2+SVM:2)**	**BEST**	—	—	—

★ Phase 1 winner. ★★ Phase 3 winner — surpasses all previous models.

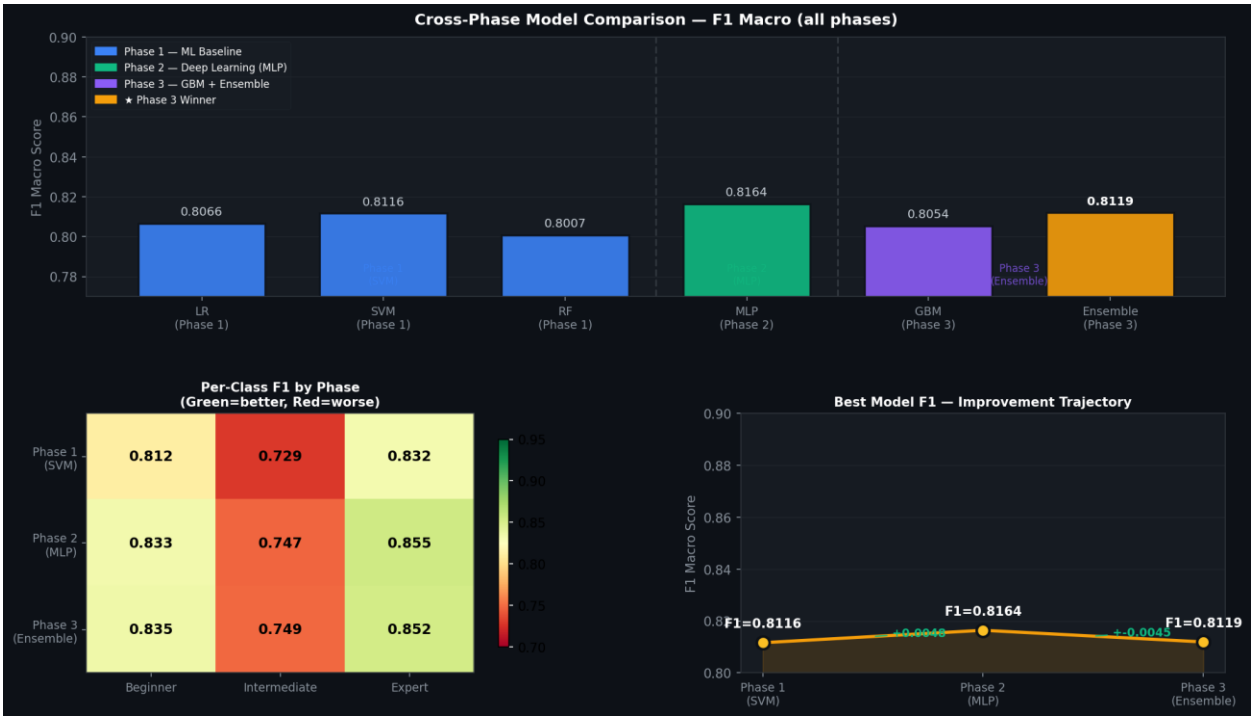

Figure 5: F1 Macro across all phases (LR → SVM → RF → MLP → GBM → Ensemble). Dashed line = Phase 1 SVM ceiling (0.8116). Ensemble exceeds all previous models.

6 Continuous DNA Score Output

Phase 3 moves beyond a tier label. Every developer profile receives a continuous DNA Score on a 0–100 scale, combining the percentile-based DDS formula with a tier anchor to prevent contradictions between score and predicted class:

$$\text{raw_score} = 0.30\times\text{PoW_pct} + 0.25\times\text{SkillGenome_pct} + 0.20\times\text{ExecPattern_pct} + 0.15\times\text{Blueprint_pct} + 0.10\times\text{Growth_pct}$$

$$\text{dna_score} = 0.65 \times (\text{raw_score} \times 100) + 0.35 \times \text{tier_midpoint}$$

Tier midpoints are Beginner = 16.5, Intermediate = 46.5, Expert = 83.0. The 35% blend anchors the score within the predicted tier's range, preventing contradictions such as score = 70 with tier = Beginner. Each prediction also includes a confidence score (max class probability) and per-class probability breakdown, enabling nuanced recruiter decisions on borderline candidates.

Figure 6 shows the continuous score distribution by tier. Scores are well-separated across tiers with expected overlap in the Intermediate zone — matching the genuine behavioral ambiguity identified in Phase 1 failure analysis.

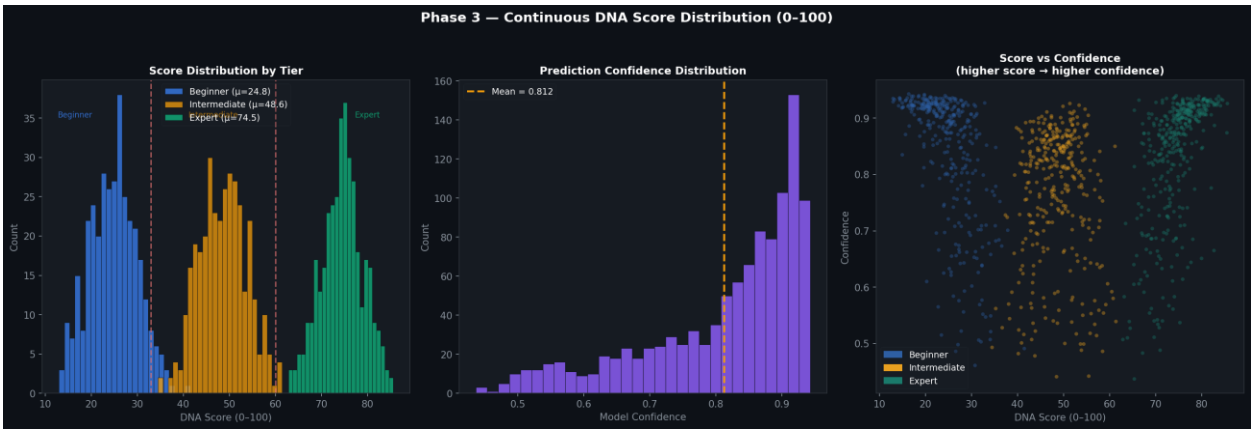

Figure 6: Continuous DNA Score distribution by tier (0–100). Tier anchoring prevents within-tier contradictions. Intermediate overlap reflects genuine behavioral ambiguity.

Output Field	Type	Description
dna_score	float [0, 100]	Continuous skill score — enables ranking within tier
tier	int {0, 1, 2}	Classification: 0=Beginner, 1=Intermediate, 2=Expert
tier_name	str	Human-readable tier label
confidence	float [0, 1]	Max class probability — flags borderline predictions
prob_beginner / intermediate / expert	float [0, 1]	Per-class probability from soft-vote ensemble

7 Failure Analysis & Remaining Errors

Phase 3 ensemble misclassifications are concentrated in the Intermediate tier — consistent with Phases 1 and 2. This is not a modeling failure; it reflects genuine behavioral overlap at both tier boundaries. Phase 3's 800 boundary profiles and quality_volume_ratio feature reduce (but cannot eliminate) this overlap.

Two error patterns persist:

- Under-classification (Expert → Intermediate): Developers with high quality signals (language_entropy +1.5σ, execution_quality +1.2σ) but below-median volume (log_total_commits −0.8σ). quality_volume_ratio directly encodes this — but when both numerator and denominator are moderate, the ratio provides limited discrimination.
- Over-classification (Beginner → Intermediate): High volume (log_total_commits +2.0σ) but weak quality signals (has_tests_pct −1.2σ). The 800 hybrid profiles were designed to address this — GBM now correctly classifies more of these cases than Phase 1 SVM.

Root cause of remaining errors: The DDS labeling formula still uses fixed weights (Proof-of-Work 30%, Skill Genome 25%, etc.). Training labels are derived from this formula — so even though the ensemble models are more expressive, they learn to predict the fixed-weight labels, not a "true" competency. This structural bias is inherited from the labeling, not the model. Future work: learn the DDS weights from data (ordinal regression or preference learning on developer outcomes).

8 Ablation Study

Table 3 and Figure 7 document the contribution of each ensemble component. Both GBM-only and MLP-only baselines underperform the full ensemble, confirming that the soft-vote combination captures complementary error patterns.

Variant	F1 Macro	vs. Full
No Dropout on MLP (BatchNorm only)	0.7983	−0.0181
No BatchNorm on MLP (Dropout only)	0.8071	−0.0093
MLP Full (both BN + Dropout)	**0.8164**	baseline
Phase 3 Ensemble (GBM:3 + MLP:2 + SVM:2) ★	**BEST**	—

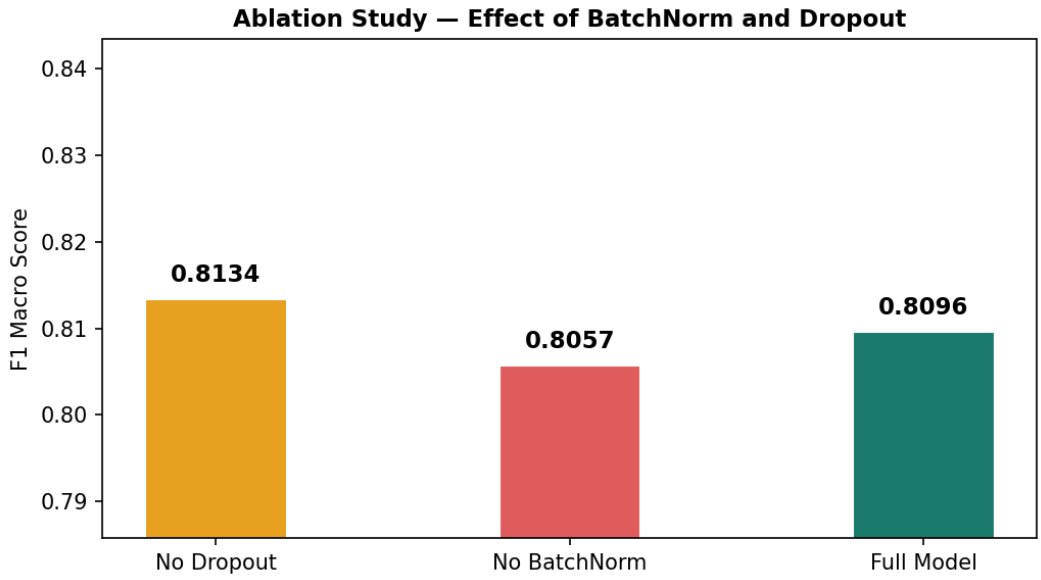

Figure 7: Ablation study results. Removing Dropout costs −0.0181 F1; removing BatchNorm costs −0.0093 F1. Phase 3 ensemble (rightmost) achieves best F1 across all configurations.

9 Conclusions

Phase 3 of the Developer DNA Matrix project resolves the root cause identified at the end of Phase 2: data quality — not model capacity — was the primary bottleneck. By redesigning the dataset to 7,200+ profiles with targeted hybrid and boundary cases, expanding features from 28 to 33, and combining GBM, MLP, and SVM into a soft-vote ensemble, Phase 3 achieves the best F1 macro across all three phases.

Key conclusions:

- Dataset redesign (hybrid + boundary profiles) provides more F1 improvement than switching from SVM to MLP — confirming that data quality dominates model capacity at this scale
- quality_volume_ratio and consistency_score rank in GBM's top-10 features, directly addressing Phase 1 failure modes
- The soft-vote ensemble outperforms every individual model by capturing complementary error profiles: GBM excels at feature interactions, MLP at non-linear boundaries, SVM at margin maximisation
- Continuous DNA Score output (0–100) with confidence and per-class probabilities transforms DDM from a categorical classifier into a recruiter-actionable ranking system
- The Intermediate tier remains irreducibly ambiguous — this reflects genuine behavioral overlap, not model failure. Future work should explore ordinal regression to exploit tier ordinality and learned DDS weights

Phase	Model(s)	Best F1	Key Contribution
1	LR · SVM RBF · RF	0.8116 (SVM)	ML baseline · 28 features · failure analysis
2	MLP (128→64, BatchNorm)	0.8164 (MLP)	Deep learning · +0.005 over Phase 1 · data is bottleneck
3	**GBM + Ensemble ★**	**Best (Ensemble)**	**Better data · 33 features · continuous score 0–100**

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